RAJALAKSHMI ENGINEERING COLLEGE

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Started on	Wednesday, 30 September 2026, 10:42 AM
State	Finished
Completed on	Wednesday, 30 September 2026, 10:43 AM
Time taken	1 min 19 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	**g**	**t**	**a**	**b**
s2	**g**	x	**t**	x	**a**	**y**	**b**

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab azb	2

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
#include <string.h>

int max(int a, int b)
{
    return (a > b) ? a : b;
}

int main()
{
    char s1[1000], s2[1000];

    scanf("%s", s1);
    scanf("%s", s2);

    int m = strlen(s1);
    int n = strlen(s2);

    int dp[m + 1][n + 1];

    // Initialize first row and first column
    for (int i = 0; i <= m; i++)
        dp[i][0] = 0;

    for (int j = 0; j <= n; j++)
        dp[0][j] = 0;

    // Fill DP table
    for (int i = 1; i <= m; i++)
    {
        for (int j = 1; j <= n; j++)
        {
            if (s1[i - 1] == s2[j - 1])
            {
                dp[i][j] = dp[i - 1][j - 1] + 1;
            }
            else
            {
                dp[i][j] = max(dp[i - 1][j], dp[i][j - 1]);
            }
        }
    }

```

```
44      printf("%d", dp[m][n]);
45
46      return 0;
47  }
```

	Input	Expected	Got	
✔	aab azb	2	2	✔
✔	ABCD ABCD	4	4	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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